

HDOC DAY-6

QUESTION 1-Area of a square Garden

Question

A square garden has side s meters. Write a C program to find the area of the garden.

Step-1 Understand the problem

Input:

Side of the square garden, s , in meters

Output:

Area of the square garden in square meters

Formula:

Area = $s \times s$

Step-2 Test Case Table

Test case	input	Expected output	Actual output	result
1	S=5	25.00	25.00	pass
2	S=10	100.00	100.00	pass
3	S=2.5	6.25	6.25	pass
4	S=1	1.00	1.00	pass
5	S=0	0.00	0.00	pass

Step-3 Algorithm

Algorithm: area of a square garden

- 1.start
- 2.declare variables $side$ and $area$
- 3.Input the side of the square garden
- 4.calculate the area using:
 $Area = side \times side$
- 5.display the area of the garden
- 6.stop

Algorithm Evaluation

The algorithm takes the side of the square garden as input and calculates its area using the formula $\text{Area} = \text{side} \times \text{side}$. The calculated results for the prepared test cases match the expected results. Hence, the algorithm is correct for the tested inputs.

Step-4 C program

```
#include <stdio.h>

int main()
{
    double side, area;

    printf("Enter the side of the square garden: ");

    scanf("%lf", &side);

    area = side * side;

    printf("Area of the garden = %.2lf square meters\n", area);

    return 0;
}
```

Step 5: Compilation, Execution and Evaluation

Compilation

```
cc square.c -o square
```

Execution

```
./square
```

Step 6-

```
puppie@Anvikas-MacBook-Air Desktop % pwd
/Users/puppie/Desktop
puppie@Anvikas-MacBook-Air Desktop % nano square.c
puppie@Anvikas-MacBook-Air Desktop % ls
6                                sample1.c.save                                square.c
coding                           Screenshot 2026-08-30 at 1.19.09 PM.png
HDOC-Day                         Screenshot 2026-08-30 at 9.44.45 PM.png
puppie@Anvikas-MacBook-Air Desktop % square.c
zsh: command not found: square.c
puppie@Anvikas-MacBook-Air Desktop % at square.c
at: garbled time
puppie@Anvikas-MacBook-Air Desktop % cat square.c
#include <stdio.h>

int main()
{
    double side, area;

    printf("Enter the side of the square garden: ");
    scanf("%lf", &side);

    area = side * side;

    printf("Area of the garden = %.2lf square meters\n", area);

    return 0;
}
puppie@Anvikas-MacBook-Air Desktop % cc square.c -o square
puppie@Anvikas-MacBook-Air Desktop % ls
6                                sample1.c.save                                square
coding                           Screenshot 2026-08-30 at 1.19.09 PM.png            square.c
HDOC-Day                         Screenshot 2026-08-30 at 9.44.45 PM.png
puppie@Anvikas-MacBook-Air Desktop % ./square
Enter the side of the square garden: 5
Area of the garden = 25.00 square meters
puppie@Anvikas-MacBook-Air Desktop %
```

Evaluation

The program was compiled and executed successfully. It was evaluated using the test cases prepared in Step 2. The actual outputs matched the expected outputs for all the test cases; therefore, all test cases passed

Question 2- Sector : Angle, Area & Perimeter

Write a C program to find the angle, area, and perimeter of a sector when its arc length and radius are given.

Step1- problem understanding

Input:

- Radius of sector = r
- Arc length = l

Output:

- Angle of sector θ in radians
- Area of sector
- Perimeter of sector

Formulas:

- $\theta = l/r$
- Area = $\frac{1}{2} rl$
- Perimeter = $2r + l$

Step2-Test cases

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	r = 5, l = 10	Angle = 2.00 rad, Area = 25.00, Perimeter = 20.00	Angle = 2.00 rad, Area = 25.00, Perimeter = 20.00	Pass
2	r = 10, l = 5	Angle = 0.50 rad, Area = 25.00, Perimeter = 25.00	Angle = 0.50 rad, Area = 25.00, Perimeter = 25.00	Pass
3	r = 4, l = 8	Angle = 2.00 rad, Area = 16.00, Perimeter = 16.00	Angle = 2.00 rad, Area = 16.00, Perimeter = 16.00	Pass
4	r = 2.5, l = 5	Angle = 2.00 rad, Area = 6.25, Perimeter = 10.00	Angle = 2.00 rad, Area = 6.25, Perimeter = 10.00	Pass
5	r = 1, l = 1	Angle = 1.00 rad, Area = 0.50, Perimeter = 3.00	Angle = 1.00 rad, Area = 0.50, Perimeter = 3.00	Pass

Step 3: Algorithm and Evaluation

Algorithm

1. Start.
2. Declare radius, arcLength, angle, area, and perimeter.
3. Input the radius and arc length.
4. Calculate the angle using:
$$\text{angle} = \text{arcLength} / \text{radius}$$
5. Calculate the area using:
$$\text{area} = 0.5 \times \text{radius} \times \text{arcLength}$$
6. Calculate the perimeter using:
$$\text{perimeter} = 2 \times \text{radius} + \text{arcLength}$$
7. Display the angle, area, and perimeter.
8. Stop.

Algorithm Evaluation

The algorithm takes the radius and arc length as input and calculates all three required quantities using their respective formulas. The calculated values for all prepared test cases match their expected outputs; therefore, the algorithm is correct for the tested inputs.

Step 4: C Program

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    double radius, arcLength;
```

```
    double angle, area, perimeter;
```

```
    printf("Enter the radius: ");
```

```
    scanf("%lf", &radius);
```

```
printf("Enter the arc length: ");
scanf("%lf", &arcLength);

angle = arcLength / radius;
area = 0.5 * radius * arcLength;
perimeter = 2 * radius + arcLength;

printf("Angle = %.2lf radians\n", angle);
printf("Area = %.2lf square units\n", area);
printf("Perimeter = %.2lf units\n", perimeter);

return 0;

}
```

Step 5: Compilation, Execution and Evaluation

Compilation

```
cc sector.c -o sector
```

Execution

```
./sector
```

For Test Case 1:

Enter the radius: 5

Enter the arc length: 10

Angle = 2.00 radians

Area = 25.00 square units

Perimeter = 20.00 units

```
[puppie@Anvikas-MacBook-Air Desktop % pwd
/Users/puppie/Desktop
[puppie@Anvikas-MacBook-Air Desktop % mkdir HDOC-Day 6
mkdir: HDOC-Day: File exists
mkdir: 6: File exists
[puppie@Anvikas-MacBook-Air Desktop % HDOC-Day 6
zsh: command not found: HDOC-Day
[puppie@Anvikas-MacBook-Air Desktop % pwd
/Users/puppie/Desktop
[puppie@Anvikas-MacBook-Air Desktop % nano sector.c
[puppie@Anvikas-MacBook-Air Desktop % cc sector.c -o sector
[puppie@Anvikas-MacBook-Air Desktop % ./sector
Enter the radius: 5
Enter the arc length: 10
Angle = 2.00 radians
Area = 25.00 square units
Perimeter = 20.00 units
puppie@Anvikas-MacBook-Air Desktop %
```

Evaluation

The program was compiled and executed successfully. The program was tested using the test cases prepared in Step 2. The actual outputs matched the expected outputs for all test cases; therefore, all test cases passed.

